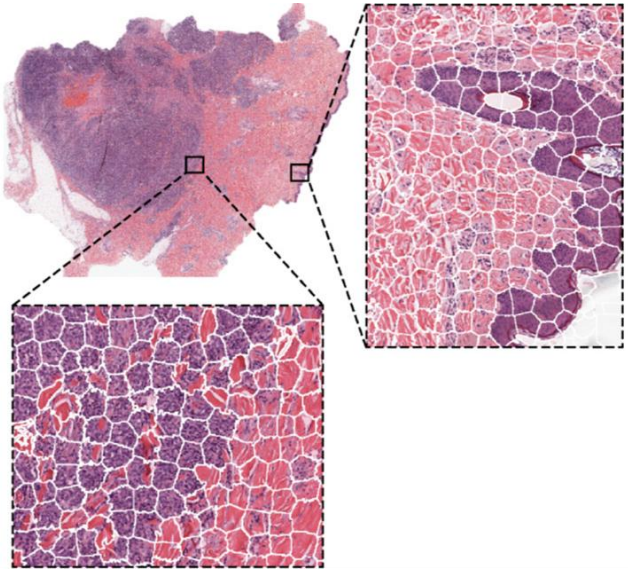
A Whole Slide Encoder for Pathology

Ali Kerem Bozkurt
Eren Berke Demirbaş
Serra Yılmaz

Whole Slide Imaging (WSI)

- Special imaging for pathology
- Images are at the gigapixel level
- Resolution can be up to 100,000x100,000
- With regular ViT patching (14), number of patches can be up to 50 million!
- We need a multi-stage design

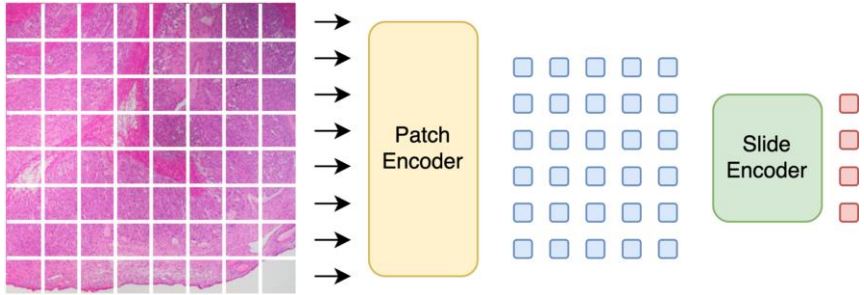


Common Approach

- 2-stage design

- 1. Stage
  - Divide into a grid of patches of manageable resolution (e.g. 512x512)
  - Encode these patches with a pretrained patch encoder specialized in pathology

- 2. Stage
  - Use a slide encoder to summarize the semantic information of the WSI



Our Approach

1 - Dataset

- Dealing with TCGA data is time consuming
- Skipped this stage and use embeddings from HistGen[1] dataset
- HistGen has pre-extracted DINOv2[2] embeddings for ~7800 WSIs
- Also has refined report pairs & coordinate information
- We used ~5000 for training, ~1400 for testing

HistGen: Histopathology Report Generation via Local-Global Feature Encoding and Cross-modal Context Interaction

Zhengrui Guo[1], Jiabo Ma[1], Yingxue Xu[1], Yihui Wang[1], Liansheng Wang[3], and Hao Chen[1,2](✉)

[1]Department of Computer Science and Engineering, The Hong Kong University of Science and Technology
{zguobc, jmabq, yxueb, ywangrm}@connect.ust.hk, jhc@cse.ust.hk
[2]Department of Chemical and Biological Engineering, The Hong Kong University of Science and Technology
[3]School of Information Science and Engineering, Xiamen University
lswang@xmu.edu.cn

[1] Guo et al. 2024
[2] Oquab et al. 2023

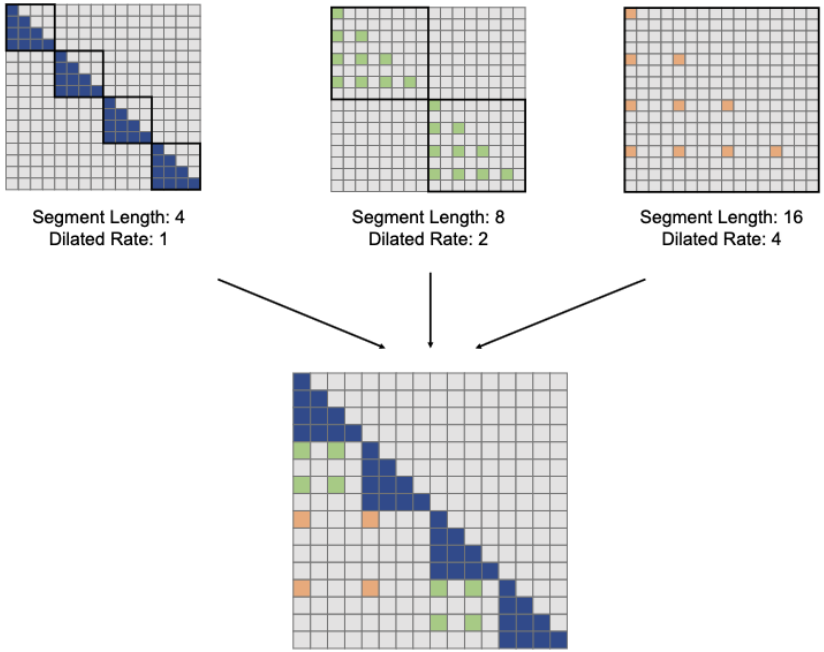
Our Approach
2 - Vision Encoder

- LongNet[3] : Scaling Transformers to a Billion Tokens

- Architecture Details
  - 6 Layers
  - 1024 Embedding dimension
  - 4096 FFN embedding dimension
  - 16 Heads

- ~76M parameters

- Added 2D sin-cos pos embeds[4]

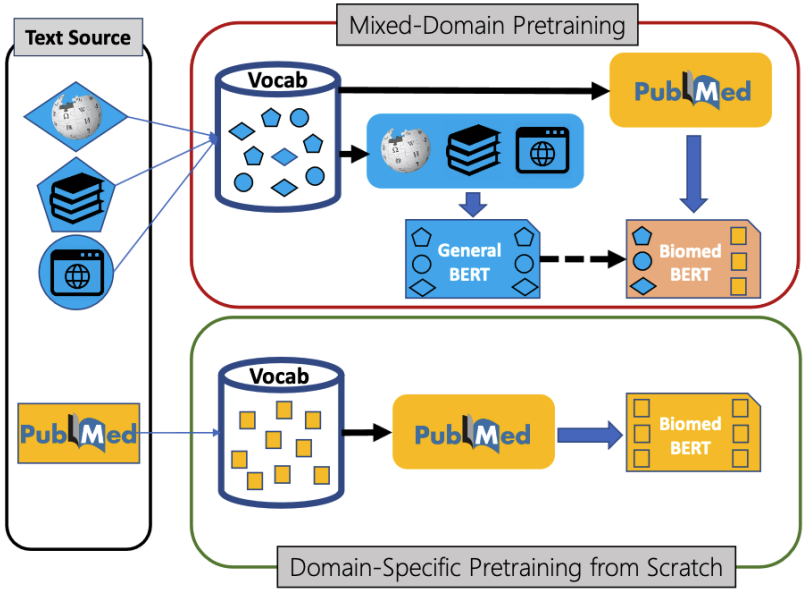


Dilated Attention

[3] Ding et al. 2023
[4] Vaswani et al. 2017

Our Approach
3 - Text Encoder

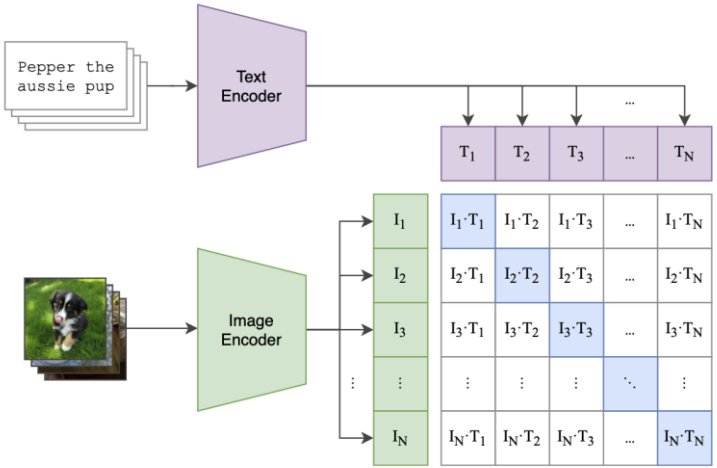
- PubMedBERT[5]
- Pretrained on abstracts of papers from PubMed
- Added a linear layer to the end to match the dimensions
- Only the projector was trained



[5] GU et al. 2021

Our Approach
4 - Training Objective

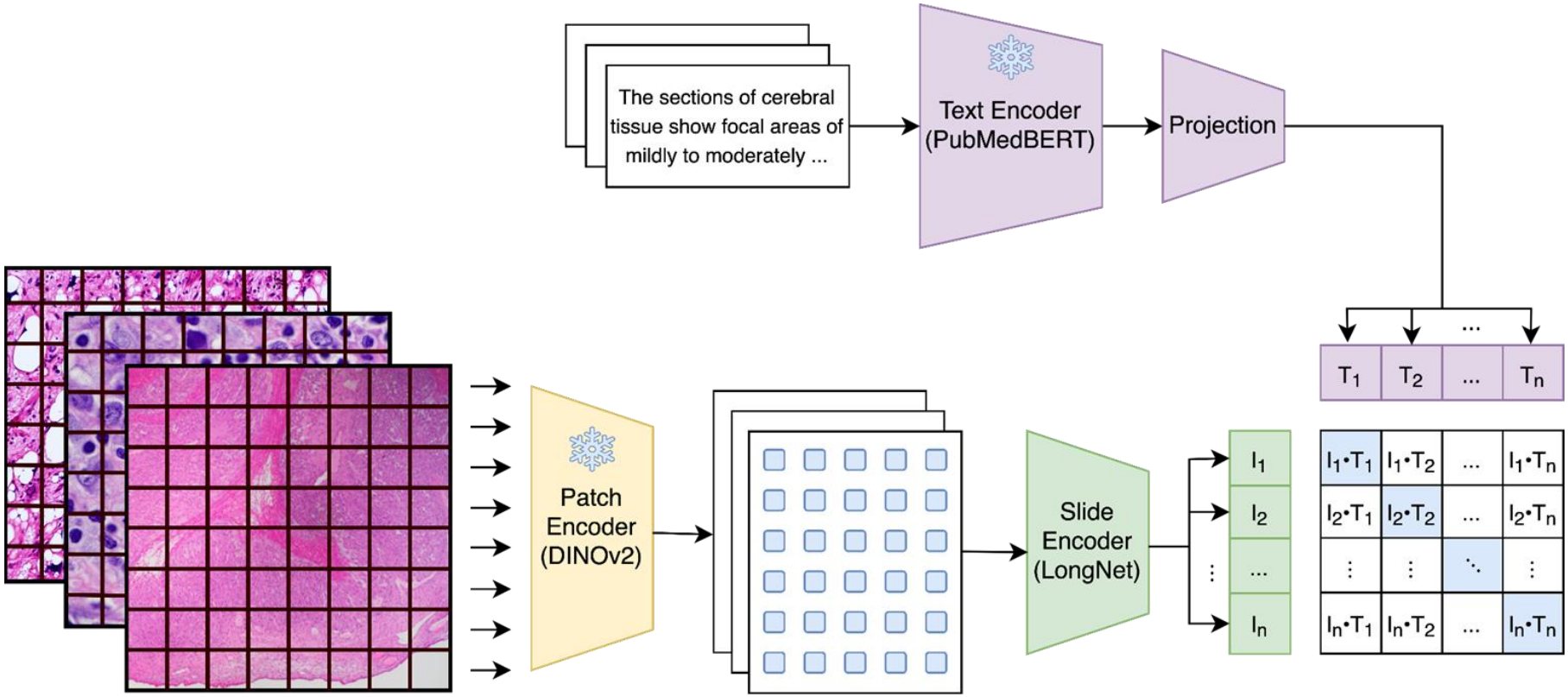
- Multimodal training for WSI-text pairs
- CLIP[6] (Contrastive Language-Image Pretraining)
- We will use an open source implementation: OpenCLIP[7]



Contrastive Language-Image Pretraining

[6] Radford et al. 2021
[7] Cherti et al. 2024

Diagram


The sections of cerebral tissue show focal areas of mildly to moderately ...
Text Encoder (PubMedBERT)
Projection
T1
T2
...
Tn
Patch Encoder (DINOv2)
Slide Encoder (LongNet)
I1
I2
...
In
I1•T1
I1•T2
I1•Tn
I2•T1
I2•T2
I2•Tn
In•T1
In•T2
In•Tn

Development Stages

1 - Dataset Preprocessing

- Downloaded the dataset from Huggingface
  - Features, h5 files, report data
  - +300 GB

- Converted coordinates from h5 files to positions
  - Position = (coord - min coord) / patch size
  - Can have fractional

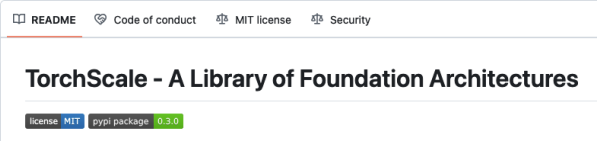
- Added custom dataset class and functions
  - Collate function handles padding in batches
  - Returns padding masks

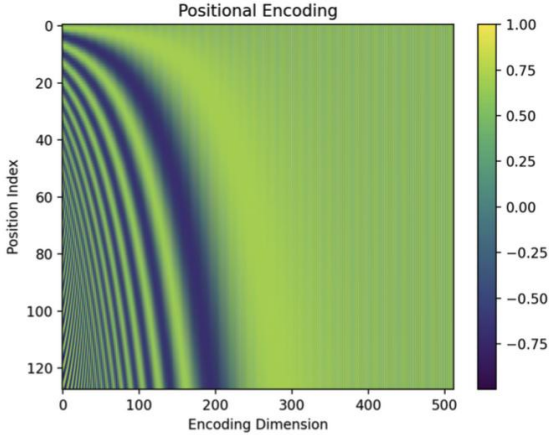
Development Stages

2 - Vision Encoder Implementation

- LongNet from TorchScale
  - For language modeling
  - Follows modern LLM implementations

- Modifications
  - Discarded embedding layer
  - Added 2D sin-cos positional encodings
  - Added CLS token
  - Precision casting
  - LayerNormFp32 for numerical stability





Sin-cos positional encoding[8]

[8] KiKaBeN 2021

Development Stages

3 - Text Encoder Implementation

- Custom HFTextEncoder
  - Added projection layer
  - PubMedBERT is frozen
  - Precision casting

4 - Training Pipeline Modifications

- Added the new modules
  - Dataset
  - Vision encoder
  - Text encoder

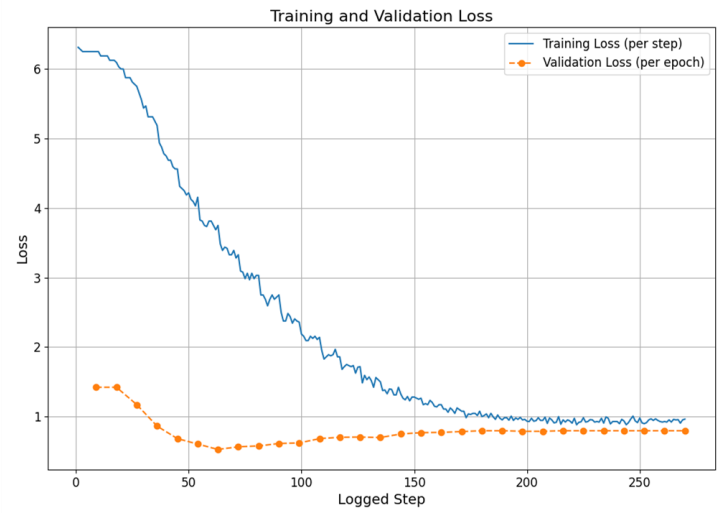
- Handle coordinates and paddings

Training Details

- Learning Rate: 2e-4
- %10 Warm up steps
- Batch size: 4
- Gradient Accumulation: 16
- AdamW Optimizer
  - Weight decay: 0.1
- Gradient Checkpointing
- BF16 precision
- Scheduler: Cosine
- 30 Epochs
- Loss : InfoNCE (CLIP)

Results

(Test set has ~1400 samples!)



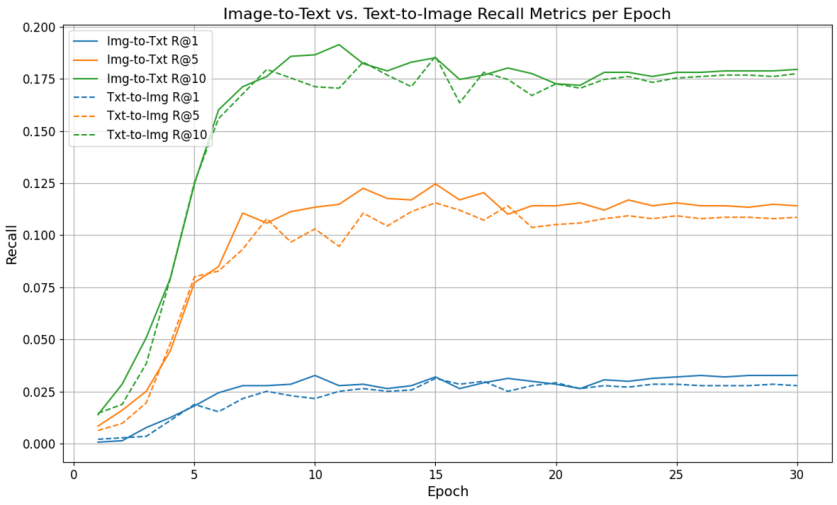


Image-to-text:

- Median rank: 59
- R@1 : 0.0327
- R@5 : 0.1246
- R@10 : 0.1914

Text-to-image

- Median rank: 60
- R@1 : 0.0313
- R@5 : 0.1155
- R@10: 0.1858

Results

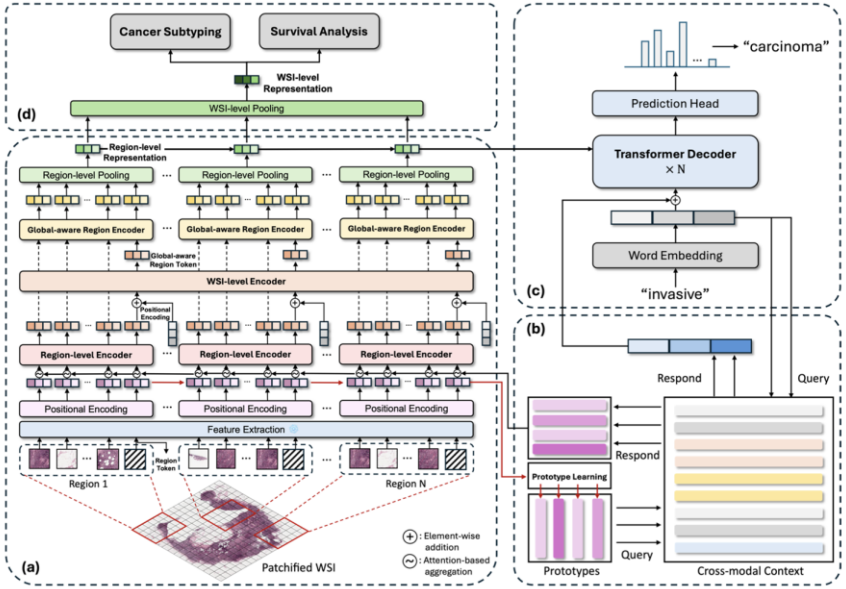


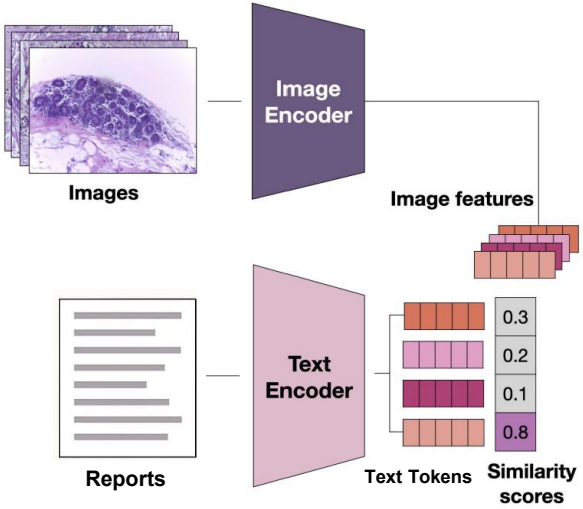
Fig. 1. Overview of the proposed HistGen framework: (a) local-global hierarchical encoder module, (b) cross-modal context module, (c) decoder module, (d) transfer learning strategy for cancer diagnosis and prognosis.

Method	R@1	R@5	R@10
Image-to-Text			
Ours	0.0327	0.1246	0.1914
Baseline (HistGen)	**0.11**	**0.25**	**0.38**
Text-to-Image			
Ours	0.0313	0.1155	0.1858
Baseline (HistGen)	**0.10**	**0.27**	**0.37**

~537M #parameters !

Zero-Shot Classification

1. Image Embedding
    a. Pre-extracted patch features+coords
    b. Fine-tuned LongNet Image Encoder
    c. Output: Single normalized image embedding per slide
2. Report Embedding
    a. Tokenize clinical reports with Pre-trained PubMedBERT Text Encoder
    b. Output: Single normalized text embedding per report
3. Zero-Shot Classification
    a. Build class prototypes from each disease name (32 classes)
    b. Compute cosine sims: image→class and report→class
    c. Fusion (α = 0.5)
        i. Score = (1-α).s_img + α.s_text
    d. Argmax over scores for final prediction



Lu et al., 2023

Results

Only Image

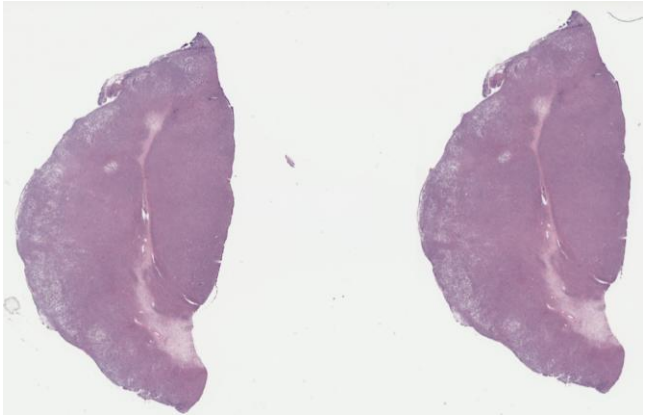
Overall Acc: 34.52% (496/1437)

- Uveal Melanoma (UVM) : 92.86% (13/14)
- Pheochromocytoma and Paraganglioma (PCPG) : 80.77% (21/26)
- Skin Cutaneous Melanoma (SKCM) : 76.92% (10/13)
- Lymphoid Neoplasm Diffuse Large B-Cell Lymphoma (DLBC) : 75.00% (6/8)
- Glioblastoma Multiforme (GBM) : 73.68% (14/19)

Image+Report

Overall Acc: 29.16% (419/1437)

- Pheochromocytoma and Paraganglioma (PCPG): 84.62% (22/26)
- Glioblastoma Multiforme (GBM) : 78.95% (15/19)
- Uveal Melanoma (UVM) : 78.57% (11/14)
- Lymphoid Neoplasm Diffuse Large B-Cell Lymphoma (DLBC) : 75.00% (6/8)
- Prostate Adenocarcinoma (PRAD) : 66.67% (48/72)



The WSI for sample TCGA-G8-6914-01Z-00-DX1 was obtained from the Genomic Data Commons Data Portal (National Cancer Institute, n.d.).

Slide ID: TCGA-G8-6914-01Z-00-DX1.910a33a3-3c27-4368-a223-d15a5e024c5a

Report: 'The report describes two specimens: a left cervical lymph node biopsy and bone marrow biopsies of both the right and left iliac crests. The left cervical lymph node biopsy is diagnosed as malignant lymphoma, specifically a diffuse large cell type that exhibits B-cell phenotype. A preliminary flow cytometric report shows that the lymphoma cells are positive for CD19, CD20, and CD22. The bone marrow biopsies indicate orderly hematopoiesis with no presence of lymphoma. The biopsy data shows normal hematopoiesis with no dysplasia observed in erythroid or myeloid series. Iron stores were not observed, and there was no abnormal infiltrate. Megakaryocytes were rare, and none were observed in the cervical level 5 lymph node biopsy.'

Project: 'lymphoid neoplasm diffuse large b-cell lymphoma'

Project Code: 'DLBC'

References

Ding, J., Ma, S., Dong, L., Zhang, X., Huang, S., Wang, W., … & Zhou, M. (2023). LongNet: Scaling Transformers to 1,000,000,000 Tokens. *arXiv preprint arXiv:2307.02486*. https://arxiv.org/abs/2307.02486

Ding, T., Wagner, S. J., Song, A. H., Chen, R. J., Lu, M. Y., Zhang, A., … & Mahmood, F. (2024). Multimodal Whole Slide Foundation Model for Pathology. *arXiv preprint arXiv:2411.19666*. https://arxiv.org/abs/2411.19666

Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., … & Poon, H. (2020). Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing. *arXiv preprint arXiv:2007.15779*. https://arxiv.org/abs/2007.15779

Guo, Z., Ma, J., Xu, Y., Wang, Y., Wang, L., & Chen, H. (2024). HistGen: Histopathology Report Generation via Local-Global Feature Encoding and Cross-modal Context Interaction. *arXiv preprint arXiv:2403.05396*. https://arxiv.org/abs/2403.05396

Ilharco, G., Wortsman, M., Wightman, R., Gordon, C., Carlini, N., Taori, R., … & Zettlemoyer, L. (2021). OpenCLIP: An Open Source Implementation of CLIP. *GitHub repository*. https://github.com/mlfoundations/open_clip

KiKaBeN. (2021). Transformer's Positional Encoding: How Does It Know Word Positions Without Recurrence? Published October 30, 2021. https://kikaben.com/transformers-positional-encoding/

Lu, M. Y., Chen, B., Zhang, A., Williamson, D. F. K., Chen, R. J., Ding, T., Le, L. P., Chuang, Y. S., & Mahmood, F. (2023). *Visual Language Pretrained Multiple Instance Zero-Shot Transfer for Histopathology Images* (arXiv:2306.07831). arXiv.

National Cancer Institute. (n.d.). *TCGA-G8-6914-01Z-00-DX1 slide image file (UUID 62a3ed21-180b-4f25-b29d-8c88cda7a65d)*. Genomic Data Commons Data Portal. Retrieved June 10, 2025, from https://portal.gdc.cancer.gov/files/62a3ed21-180b-4f25-b29d-8c88cda7a65d

Oquab, M., Darcet, T., Moutakanni, T., Vo, H. V., Szafraniec, M., Marc, M., … & Bojanowski, P. (2023). DINOv2: Learning Robust Visual Features without Supervision. *arXiv preprint arXiv:2304.07193*. https://arxiv.org/abs/2304.07193

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., … & Sutskever, I. (2021). Learning Transferable Visual Models From Natural Language Supervision. *arXiv preprint arXiv:2103.00020*. https://arxiv.org/abs/2103.00020

Su, J., Lu, Y., Pan, S., Murtadha, A., Wen, B., & Liu, Y. (2021). RoFormer: Enhanced Transformer with Rotary Position Embedding. *arXiv preprint arXiv:2104.09864*. https://arxiv.org/abs/2104.09864

Xu, Y., Chen, R. J., Lu, M. Y., Williamson, D. F. K., & Mahmood, F. (2024). Prov-GigaPath: A Whole-Slide Foundation Model for Digital Pathology from Real-World Data. *Nature*. https://www.nature.com/articles/s41586-024-07441-w

Yang, J., Shi, J., Tan, X., Sun, J., Li, L. H., Wang, J., … & Wang, X. (2023). PathChat: Towards Large-scale Multi-modal Foundation Model for Pathology. *arXiv preprint arXiv:2312.07814*. https://arxiv.org/abs/2312.07814

Yu, J., Wang, Z., Vasudevan, V., Yeung, L., Seyedhosseini, M., & Wu, Y. (2022). CoCa: Contrastive Captioners are Image-Text Foundation Models. *arXiv preprint arXiv:2205.01917*. https://arxiv.org/abs/2205.01917

Zhou, J., Wei, C., Wang, H., Shen, W., Xie, C., Yuille, A., & Kong, T. (2021). iBOT: Image BERT Pre-Training with Online Tokenizer. *arXiv preprint arXiv:2111.07832*. https://arxiv.org/abs/2111.07832

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, *30*.

Thank you for listening

Questions?